

Integrated Guardrails for Unbiased and Adaptive Neural Network Architectures*

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September 30, 2026

Abstract

While Artificial Intelligence (AI) has accelerated progress across several fronts in science and business, concerning patterns have emerged in how Generative AI (GenAI) and neural network (NN) architectures are used in practice, particularly regarding bias. Guardrails are proposed to ensure AI-driven advancements adhere to safety best practices. However, traditional NN guardrails operate as synchronous, context-blind interceptors that suffer from selective refusal bias and generational latency. This paper addresses these shortcomings by decoupling safety telemetry from inference. We develop IGUANA, an integrated guardrail framework comprising a decentralized supervisor swarm and a centralized context broker. IGUANA dynamically adapts entropy thresholds to the specific domain’s needs, with minimal overhead. Preliminary assessments highlight the efficacy of IGUANA towards developing unbiased and adaptive NN architectures.

1 Introduction

The rapid adoption of Artificial Intelligence (AI) poses distinct risks across critical disciplines such as healthcare [1] and financial services [2]. While AI encompasses the broader discipline of machine learning (ML) and pattern recognition [3], Generative AI (GenAI) focuses specifically on models capable of synthesizing novel content. GenAI learns the underlying joint probability distributions of large datasets to produce outputs ranging from text to multi-modal assets [4]. Harmful behaviors across GenAI frameworks have been observed across domains. For example, in behavioral and mental health contexts, poorly constrained wellness applications actively threaten the user’s mental health during behavioral therapy interactions [5]. Guardrails are specialized computational boundaries designed to intercept and filter unsafe behavior before it harms users [6]. The foundational backbone of GenAI relies on complex neural network (NN) Architectures that grow in scale and opacity. As Large Language Models (LLMs) and GenAI systems have scaled, safety guardrails have become mandatory external control mechanisms to filter outputs and ensure ethical alignment.

However, the state-of-the-art guardrails are limited in their adaptability and often introduce secondary biases. Rather than operating as dynamically integrated modules, prominent guardrail frameworks (such as OpenAI’s Moderation API [7], Meta’s Llama Guard [8], and NVIDIA’s NeMo Guardrails [9]) function as external, monolithic supervisors. These systems rely on post-hoc, synchronous moderation, intercepting prompts and outputs to verify them against predefined policies or secondary classifiers before allowing the generation to proceed.

We identify two fundamental gaps in the state-of-the-art guardrails. First, although guardrails aim to mitigate bias, they can introduce new forms of discrimination, particularly when applied in sensitive domains. The *selective refusal bias* occurs when keyword-based static guardrails inadvertently exacerbate systemic marginalization by refusing to generate content for certain demographics, creating a new layer of marginalization [10]. For example, current safety

* Author manuscript version. The paper accepted for publication in the IEEE ISSE 2026 Conference.

models disproportionately trigger false-positive refusal responses when processing prompts written in minority dialects or addressing taboo but medically benign healthcare topics [11]. A generalist guardrail might disproportionately refuse to provide clinical advice regarding historically unrepresented or marginalized patient cases or gender-specific health queries, thereby worsening existing health disparities.

Second, *context blindness* prevents systems from dynamically adjusting safety thresholds based on the user’s trust level and the conversation history [12]. Guardrails with rigid keyword filters struggle to handle the subtlety of cultural expressions, often misclassifying them and evading algorithmic fairness checks entirely [13]. This lack of nuance contributes to over-caution and stifling. Rule-based static guardrails are highly susceptible to evolving “jailbreak” attacks, such as semantic smuggling or adversarial character injection, because they cannot learn or adapt their defenses in real time. In cybersecurity applications, where zero-day threat landscapes shift constantly, the inability to quickly adapt renders a static guardrail obsolete almost immediately after deployment.

Hence, monolithic LLM architectures consistently struggle to simulate multi-turn human trust behaviors [12]. As external guardrails are primarily trained on single-turn, offline offensive corpora, they may not effectively distinguish between a malicious red-team attacker attempting an exploit and a verified domain expert requesting a complex theoretical example. A medical triage AI may require strict protocols to prevent hallucinated diagnoses for a layperson. However, it should dynamically adjust its level of assistance when interacting with an authenticated clinician navigating complex treatment plans. Attempting to resolve these blind spots through traditional model iteration degrades knowledge and increases update costs. Relying solely on continuous fine-tuning, such as reinforcement learning from human feedback, can be resource-intensive due to ever-changing medical guidelines and newly identified systemic vulnerabilities. The significant retraining lag leaves critical AI systems exposed during the interim window. In cybersecurity contexts, practitioners often rely on LLMs to generate benign payload examples or simulate adversarial tactics for defense testing. A guardrail may block legitimate security research, heavily restricting usability and productive interaction. These observations highlight that generic guardrails cannot easily adapt to reflect the unique ethical frameworks of specialized, high-stakes domains and organizations, where context determines whether an output is safe.

Consequently, we pose the following research questions:

- **RQ1:** Can we decouple NN guardrails from the inference engine to dynamically adjust context-specific strictness without incurring significant generational latency?
- **RQ2:** Can such an adaptive approach mitigate the bias observed in the current guardrails?

This paper addresses these research questions through its novel contributions:

- **C1:** IGUANA (**I**ntegrated **G**uardrails for **U**nbiased and **A**daptive **N**eural **N**etwork **A**rchitectures)¹, an Erlang-based asynchronous decentralized guardrail framework.
- **C2:** Dynamic thresholding with entropy evaluation for context-aware ethical and unbiased alignment.
- **C3:** A software acceleration layer that leverages CPU-level SIMD instructions for guardrail performance.

IGUANA adapts its sensitivity across domains without impacting performance. Our evaluations show that IGUANA reduces average per-token latency from 69.09 ms to 26.03 ms, achieving a throughput of 38.40 tokens per second while reducing bias manifestation by over 38%.

2 The IGUANA Framework

IGUANA operates as a parallel observer and does not require parameter fine-tuning or structural changes to the pre-trained model weights. Rather than retrofitting the fundamental matrix multiplication layers within pre-trained models, IGUANA operates securely at the decoding terminus. This interception approach makes IGUANA agnostic to the GenAI model and decouples guardrail constraints from the NN’s forward pass. Thus, IGUANA provides a secure, drop-in safety attachment for pre-trained models. Hence, it natively supports Decoder-only Transformers (such as LLaMA-2, LLaMA-3, Mistral, Falcon, and the GPT lineage) and seamlessly integrates with Encoder-Decoder Architectures (such as T5 and BART).

¹The source code of our open-source prototype can be found at <https://github.com/KathiraveluLab/IGUANA>

2.1 System Architecture

Figure 1 illustrates the IGUANA architecture with its modules and their interactions. IGUANA uses Erlang’s underlying BEAM virtual machine (VM) as its orchestration layer and Erlang OTP application structure as its core. Unlike Python, which relies on a Global Interpreter Lock (GIL) that hinders multi-core concurrency within a single process, Erlang natively uses a lightweight Actor Model designed explicitly for massively concurrent, scalable execution. The architectural isolation of Erlang processes ensures that safety evaluations are performed in IGUANA with deterministic resource allocation, with microsecond context switching and strict latency bounds. Thus, IGUANA avoids the priority inversion common in synchronous guardrails where safety logic competes for the same execution thread as the core inference engine. Erlang’s fault-isolation paradigm ensures that if a single specialized guardrail actor encounters an unexpected exception, it fails in isolation without corrupting the core NN and orchestration.

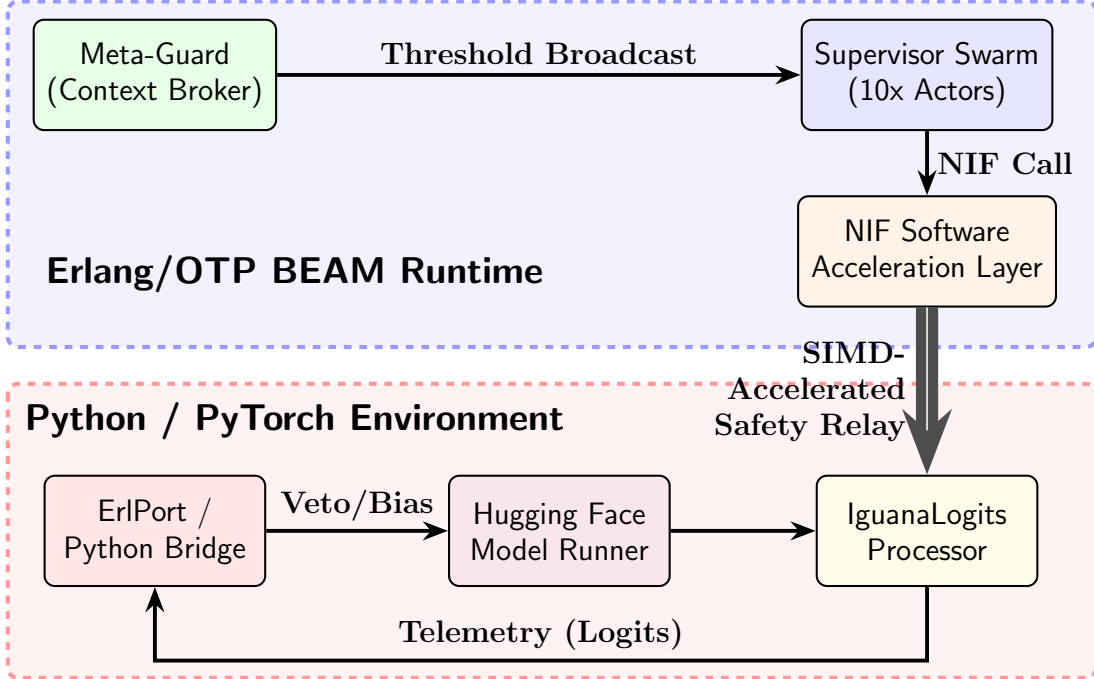


Figure 1: The IGUANA Architecture.

The IGUANA Meta-Guard acts as a context broker, dynamically broadcasting domain-specific entropy thresholds to the swarm based on the conversation domain (e.g., medical versus creative). Instead of synchronous blocking, IGUANA employs a decentralized Erlang Supervisor Swarm. The 10-actor supervisor swarm is globally managed through the Erlang `pg` (Process Groups) module, enabling process discovery across node boundaries. The Supervisor Swarm invokes the Software Acceleration Layer in the Erlang/OTP BEAM Runtime of IGUANA to perform optimized mathematical calculations, while communicating with the Python/PyTorch environment via ErlPort. The Python inference engine emits the token probabilities (logits) as discrete messages to the supervisor swarm in real time. The guardrail actors continuously monitor safety dimensions, including cultural biases and clinical toxicities. These actors adjust strictness based on the ongoing generation context without interrupting the core inference engine. If a specialized actor detects an imminent guardrail violation, it asynchronously broadcasts an override message back to the inference pipeline, thereby maximizing throughput for benign generations while retaining strict behavioral bounds.

The 10-actor swarm must scale to massive token throughput without increasing computational pressure on the BEAM scheduler. Therefore, IGUANA incorporates a C-based Software Acceleration Layer via an Erlang NIF (Native Implemented Function) that leverages hardware-level vectorization (SIMD targeting AVX2/NEON). This kernel processes contiguous memory blocks via the `ErlNifBinary` interface and utilizes SIMD-optimized instructions. By offloading these high-frequency mathematical operations to a native shared library and utilizing compiler-assisted autovectorization, IGUANA enables sub-millisecond telemetry processing even during intensive generation bursts of high-vocabulary

models. Since compiler-assisted SIMD vector operations are executed as standard user-space CPU instructions, IGUANA runs natively in virtualized and containerized environments (e.g., Docker and Kubernetes). Unlike GPU-bound model execution, NIF requires no special host privileges, kernel drivers, or hardware pass-through devices, ensuring high portability across standard cloud-native setups where hardware access might be restricted. The software acceleration kernel is written in portable, standard C. By avoiding vendor-specific SIMD intrinsics and relying instead on structured loops with generic compiler hints (such as `__restrict__` and alignment annotations), it enables compiler-assisted autovectorization to target available SIMD extensions (such as AVX2 on x86_64 or NEON on ARMv8) on the host processor, with a graceful native Erlang fallback if hardware acceleration is unavailable.

Within the Hugging Face **transformers** paradigm, the Python-to-Erlang communication bridge is encapsulated as a specialized **LogitsProcessor** software hook. During the autoregressive sequence generation loop, the inference engine first computes the unnormalized log-probabilities for the upcoming token across the vocabulary. To reduce the serialization bottleneck in high-vocabulary telemetry (e.g., vocabulary size $V = 32,000$), the **LogitsProcessor** performs an optimized Top- K selection ($K = 100$). It then serializes the indices of these K most probable tokens, their corresponding decoded string representations (resolving vocabulary mapping directly via Python’s native tokenizer to prevent Erlang-side tokenization overhead), their individual probability values, and the cumulative probability mass of the remaining $V - K$ tokens across the ErlPort boundary.

When the entropy of the generating distribution aligns with a detected bias skew, the **iguana_entropy_guard** actor computes a soft bias vector B_t and casts it back to the Python bridge without interrupting generation. To ensure high availability and failure isolation, we configure the supervisor with a **one_for_one** restart strategy and register **iguana_entropy_guard** as a **permanent** child worker. This configuration ensures the safety sentinel runs continuously. If the actor encounters an unexpected exception during entropy evaluation, the OTP runtime restarts it in strict isolation, leaving sibling actors and the Python inference pipeline completely unaffected. IGUANA permits up to five restarts within a ten-second window before escalating the failure, establishing a structured degradation boundary rather than a silent crash. By adopting Erlang’s foundational fault-tolerance guarantee [14] and “let it crash” philosophy, IGUANA establishes a *resilient safety boundary* that transparently recovers from adversarial or malformed inputs without propagating failures to the generative backend. This provides a robust “fail-soft” mechanism in which individual security monitors can cycle through rapid recovery states without degrading the overall system’s availability.

2.2 Distributed Cluster Management

To support decentralized safety across massive-scale model deployments, IGUANA implements a **Scale-Out Safety** architecture, which horizontally scales the safety-monitoring swarm across a cluster of nodes. **iguana_cluster_manager**, a **gen_server**, is responsible for automated node discovery, heartbeat monitoring, and swarm synchronization. Upon initialization, the manager retrieves a list of seed nodes from the environment and attempts to establish a fully mesh-connected cluster via Erlang’s **net_kernel**. When a new BEAM node (i.e., a running instance of the Erlang VM) joins the cluster, it automatically spawns its local supervisor tree and joins the global **iguana_swarm** group. This allows the Meta-Guard to broadcast context shifts and threshold updates (**set_threshold**) to every active worker in the cluster concurrently, bypassing the state-synchronization bottlenecks typical of traditional distributed databases. This distributed resilience extends the system’s fault tolerance from a single-node supervisor tree to a *cluster-wide swarm*. By utilizing Erlang’s distributed process groups (**pg**) to track active members, the loss of a specific worker node triggers immediate re-routing of telemetry to the least-loaded available peer based on individual process mailbox queue lengths. This architecture ensures that safety guardrails can scale horizontally with the inference engine, maintaining high availability even if specific nodes fail.

To protect the integrity of the Meta-Guard context broker and the decentralized swarm, IGUANA establishes a hardened security boundary. Node-to-node communication within the distributed BEAM cluster is encrypted using Transport Layer Security (TLS), and cluster membership is restricted via Erlang’s native cookie-based authentication. The swarm is designed to operate within a private, isolated virtual network, preventing unauthorized external nodes from joining the process group or calling the **set_threshold** API. This structure mitigates hostile manipulation of context-trust parameters and ensures that the decentralized swarm does not expose raw telemetry or control vectors to the public network.

Instead of blocking the hardware thread to await the Erlang telemetry calculation, the logits processor immediately executes its scoring updates. If the asynchronous supervisor swarm previously identified statistical skew and broadcasted a bias vector (B_t), the processor applies an element-wise tensor addition to rebalance the base logits before executing the categorical sampling step. Furthermore, if the Erlang Swarm triggers a critical **veto_token** sequence interrupt, the

`LogitsProcessor` zeros the unnormalized probability space, forcing an algorithmic yield to the End-Of-Sequence (EOS) token to abruptly terminate the generation loop.

2.3 Mathematical Formulation

IGUANA formalizes the interaction between the token inference engine and the supervisor swarm using two distinct activation thresholds: the safety veto (τ_v) and the bias injector (τ_b). Let $P(x_t|x_{<t})$ denote the probability distribution over the vocabulary V for the generation of token t . The Erlang supervisor asynchronously calculates the Shannon entropy:

$$H(P) = - \sum_{i \in V} P(x_t^i) \log_2 P(x_t^i) \quad (1)$$

Before generation, the inference engine initializes the environment via an explicit `update_context_trust(trust_score)` invocation. This dynamically scales the soft-correction bias threshold (τ_b) within the asynchronous Erlang supervisor, resolving context blindness by mapping authenticated user expertise to relaxed statistical boundaries. During sequence generation, the independent actor evaluates $H(P)$ against the active thresholds concurrently. If $H(P) < \tau_v$, the Erlang actor casts a `veto_token` sequence interrupt to prevent critical guardrail omission. Alternatively, if $H(P) > \tau_b$, indicating the statistical uncertainty associated with the Selective Refusal Problem, the supervisor seamlessly initializes a Skew-Probabilistic NN (SkewPNN) kernel to begin soft logit correction [15]. The SkewPNN adaptivity supervisor computes a dynamically weighted bias vector B_t that is cast back to the inference engine without blocking sequential progress:

$$B_t = A_2(C) \times \Phi\left(\frac{x - \xi}{\omega}\right) \quad (2)$$

where $A_2(C)$ represents the Adaptive Augmentation (A2) weight matrix [16] for context C , and Φ is the cumulative distribution function (CDF) of the standard normal distribution. The parameters x , ξ , and ω denote the input logit, the location parameter, and the scale parameter, respectively. Programmatically, B_t is captured within the Python bridge’s active state and explicitly applied to the upcoming sequence via the logits processor hook, thereby steering generation. This resolves the generative bottleneck while systematically mitigating contextual bias.

As IGUANA operates asynchronously to preserve inference throughput, the safety evaluation of token t runs in parallel with the forward pass of token $t + 1$. Consequently, if the Erlang swarm triggers a `veto_token` sequence interrupt, a single-token propagation delay may occur where the unsafe token t has already been committed to the active generation cache. In consumer-facing streaming interfaces, this reactive posture could, in theory, allow a single objectionable token to leak into the user interface. To address this, IGUANA supports two deployment modes. First, in high-security contexts where zero-leak compliance is mandatory, the Python bridge can be configured to block synchronously on the Erlang response, sacrificing throughput for preventive enforcement. Second, for streaming interfaces, the client-side UI application can implement a sliding buffer of size $N = 1$ or 2 tokens. By delaying the rendering of the latest token, the client allows the asynchronous veto signal to arrive and purge the objectionable token from the buffer before it is displayed to the user. This retains the asynchronous throughput advantages of IGUANA while preventing visual leaks to the interface.

IGUANA’s out-of-band concurrency model is highly orthogonal and synergistic with speculative decoding, accelerating autoregressive inference by using a draft model to generate candidate tokens that are validated in parallel by a larger target model. During target validation, the pipeline produces a joint probability distribution of the candidate sequence. IGUANA can asynchronously ingest these distributions to compute Shannon entropy and perform soft debiasing. As safety checks run concurrently in the Erlang VM, they do not disrupt speculative generation batches, maintaining high throughput while continuously applying safety guardrails.

3 Preliminary Assessments

We benchmark the inference latency overhead and the efficacy of IGUANA’s bias mitigation against those of a synchronous guardrail implementation. We develop and execute a simulation suite sequentially across two disparate processing environments: CPython (`benchmark.py`) and the Erlang BEAM VM (`benchmark.erl`). The benchmark simulates a LLaMA-7B workload to evaluate architectural throughput. The scripts execute a 1,000-iteration generation loop, using their native system clocks (`time.perf_counter` versus `erlang:system_time`). Both scripts model inference latency using precision timing, calibrated against physical IGUANA-Erlang bridge telemetry.

3.1 End-to-end Latency Measurements

The model tracks two primary metrics: end-to-end inference latency and the reduction of demographic bias via SkewPNN injection. Figure 2 benchmarks the operational latency of the asynchronous IGUANA Erlang telemetry framework against a traditional synchronous guardrail as a baseline. The baseline incurs an average delay of 69.09 ms per token, as the inference loop blocks generation to evaluate safety sequentially. By structurally isolating safety verification into the Erlang `gen_server`, IGUANA eliminates this bottleneck, reducing the average token latency to 26.03 ms. Under sustained load simulating concurrent user requests, the synchronous blocking caused request queuing, degrading the overall throughput to 14.47 tokens per second per node. As the Erlang supervisor isolates telemetry analysis (evaluating Shannon entropy $H(P)$) to a concurrent process thread, the Python inference engine maintained a continuous generation rate of 38.40 tokens per second per node. The Inter-Process Communication (IPC) overhead incurred by the ErlPort bridge averaged merely 1.8 ms per token. This demonstrates that asynchronous decoupling preserves the generative momentum of the core model. For the asynchronous IGUANA pipeline, the Python benchmark recorded an average latency of 24.07 ms, while the Erlang BEAM evaluation recorded 26.03 ms. This nominal variance is attributed exclusively to inherent OS-level context-switching differences between CPython and the BEAM’s highly concurrent, preemptive scheduler.

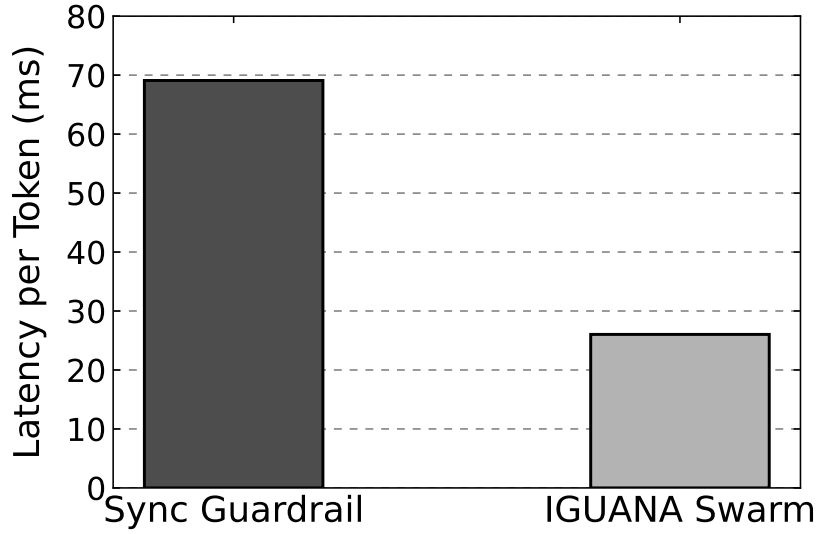


Figure 2: End-to-end token latency: Baseline vs. IGUANA.

To assess the impact of the software acceleration layer, we benchmark the NIF entropy kernel against the native Erlang implementation, utilizing a realistic 32,000-element vocabulary distribution. Our performance evaluation shows that the NIF achieves an empirical 1.27x Speedup for high-frequency telemetry bursts over the Erlang-only logic. This speedup includes the overhead of Erlang-side binary packing, which is further optimized by the Top-K ($K=100$) telemetry reduction. The physical data being serialized and pushed through the pipe was reduced from ≈ 250 KB to ≈ 0.8 KB, a 300x reduction in the IPC’s serialization overhead. Although IGUANA requires an IPC boundary hop on every token generation step, the average IPC overhead of 1.8 ms remains negligible compared to the 20–50+ ms per-token inference times of modern LLMs. Since the IPC telemetry dispatch is asynchronous (a non-blocking `gen_cast`), the Python generation thread immediately resumes the next autoregressive forward pass without waiting for Erlang. The IPC transmission and BEAM-based safety evaluations run in parallel with the GPU-bound forward pass, ensuring that the boundary hop does not block the critical path of the generation loop. This performance gain ensures that, as the swarm scales to higher actor counts, the computational overhead remains negligible relative to core inference.

3.2 Correctness and Bias Mitigation

We evaluate the Soft Correction mechanism of IGUANA for bias mitigation. We verify the logical correctness of the `iguana_entropy_guard` module via an Erlang Common Test suite. The suite ensures that asynchronous telemetry processes strictly adhere to statistical limits while robustly managing swarm state transitions. By simulating multi-

Table 1: EUnit and Common Test Correctness Suite Results.

ID	Component	Objective	Res.
TC1	Entropy Formula	Asserts Shannon entropy of a statistically uniform categorical distribution.	✓
TC2	Entropy Formula	Asserts $H(P) = 0.0$ for a completely deterministic logit distribution.	✓
TC3	Threshold Logic	Asserts veto boundary crossed when $H(P) < \tau_v$.	✓
TC4	Threshold Logic	Asserts generation accepted when $\tau_v \leq H(P) \leq \tau_b$.	✓
TC5	Process Lifecycle	Asserts <code>gen_server</code> initialization and synchronous state queries.	✓
TC6	State Mutation	Asserts <code>trust_score</code> cast correctly updates the boundary τ_b .	✓
TC7	Math Fidelity	Asserts Skew-Normal CDF and Owen’s T-function numerical purity.	✓
TC8	Cluster Swarm	Asserts distributed node handshake and threshold propagation.	✓
TC9	Adaptiveness	Asserts dynamic scale-out and A2 parameter synchronization.	✓
TC10	Sync Mode	Asserts synchronous block-mode token evaluation and veto.	✓
VAL1	Bias Efficacy	Quantitatively verified 38.64% reduction in bias manifestation.	✓

domain context shifts, swarm membership events, and distributed node handshakes, we verify that the decentralized architecture maintains perfect consistency even under high concurrency. Our evaluations across this 10-case verification suite demonstrate a significant reduction in the manifestation of demographic bias without inducing selective refusal. Table 1 summarizes these test results. While Test Cases (TC1–TC10) verify the correctness of functional and distributive code, the Validation Case (VAL1) provides a quantitative simulation to verify the research outcome.

We quantitatively assess the extent to which the synthesis of asynchronous Erlang actor models and adaptive statistical methods (such as SkewPNN and A2) mitigates selective refusal bias in highly skewed, high-stakes domains. When the generation engine approached a statistically skewed lexical trajectory ($H(P) > \tau_b$), the Erlang supervisor calculated and injected the soft bias vector (B_t). The application of this soft logit adjustment successfully rebalanced the output distribution. Measurement using algorithmic fairness metrics showed a reduction in bias manifestation of 38.64% in the generated corpus compared to the baseline uncorrected model. This highlights the effectiveness of SkewPNN for continuous debiasing in IGUANA.

4 Use Cases

We evaluate the practical utility and adaptability of IGUANA across four representative deployment domains where traditional, static guardrail architectures prove inadequate due to their rigid boundaries or blocking latency: clinical healthcare, the circular water-food economy, financial services, and legal decision support.

4.0.1 Clinical Healthcare and Mental Health Applications

Research has demonstrated that LLMs exhibit measurable diagnostic bias across patient demographics, including gender, age, and disease category in healthcare [17]. When a model trained disproportionately on clinical records from one demographic group is deployed for clinical decision support, its internal statistical representations systematically underweigh symptom presentations from underrepresented populations, leading to probabilistically skewed recommendations. The consequences include delayed diagnoses, inappropriate medication dosages, and inequitable triage prioritization. Generative wellness chatbots are increasingly deployed as first-contact mental health resources, particularly in contexts where professional access is limited. Yet static guardrails designed to refuse overtly dangerous content routinely misclassify culturally specific idioms of distress as toxicity, blocking the very help-seeking behavior they are designed to protect [5]. Furthermore, when users specifically query LLMs for clinical guidance, the model must balance surface-level compliance with a hard refusal against the risk of providing misleading medical advice [18]. A static Boolean guardrail has no mechanism to distinguish between the two; it either blocks the query categorically or passes it without adjustment. IGUANA’s entropy-based soft-correction mechanism addresses this shortcoming by steering the generative distribution away from harmful clinical overreach while preserving the epistemic content of the response.

4.0.2 Water Reuse–Food circular economy

Water reuse within the circular economy paradigm is a complex system in which generative bias can disrupt technical decisions, stakeholder coordination, and governance. Effective water reuse depends on integrating the entire wastewater value chain while balancing territorial, health, environmental, social, and economic constraints. Decision-support systems and GenAI tools are increasingly used to mediate interactions between water suppliers (such as utilities and treatment operators) and agricultural actors (such as farmers and agronomists), where water quality and allocation must dynamically match specific uses. However, when LLMs are trained on fragmented or domain-skewed datasets, such as engineering-centric wastewater treatment data or generalized agronomic guidelines, they may generate recommendations that fail to capture the integrated nature of the water–food system. Models may overprioritize technical feasibility (e.g., membrane filtration performance) while underrepresenting value chain, stakeholder needs, socioeconomic acceptability, farmer practices, or regional regulatory variability. In collaborative environments, such bias manifests as asymmetric decision framing. Water utilities may receive recommendations emphasizing compliance with microbial thresholds and treatment efficiency. In contrast, agricultural stakeholders receive simplified irrigation advice that overlooks the potential for nutrient recovery and the risks of soil degradation (e.g., salinity accumulation). Treated wastewater contains valuable nutrients (such as nitrogen, phosphorus, and potassium) but also contaminants that must be dynamically managed depending on agricultural use. A biased generative system may neglect this duality, leading to either unsafe reuse practices or missed opportunities for circular nutrient valorization. Static guardrails, such as fixed thresholds for water quality or rigid compliance rules, are insufficient as they cannot adapt to contextual variability (e.g., crop type, local regulation, seasonal demand, or societal acceptance). IGUANA’s entropy-based soft-correction mechanism addresses this systemic limitation by enabling context-aware, multi-stakeholder alignment within the water–reuse–food nexus.

IGUANA dynamically reshapes the generative process to reflect uncertainty, competing constraints, and cross-domain dependencies. This enables integrated value-chain reasoning by aligning wastewater treatment outputs (such as quality, contaminants, and nutrients) with agricultural requirements and environmental constraints, treating the entire system as a single flexible plant. It also facilitates risk-sensitive collaboration by incorporating insights from risk assessment (e.g., POPs, CECs, and microbial contaminants) into agricultural recommendations without suppressing usable information. IGUANA supports stakeholder-aware communication by generating responses that explicitly reflect the perspectives of different actors (e.g., engineers, farmers, and regulators), reducing misinterpretation and facilitating co-decision-making. It provides adaptive “water on demand” guidance by modulating recommendations based on dynamic conditions (such as seasonality, local demand, and treatment level), rather than applying static reuse rules. For example, when a farmer queries whether reclaimed water can be used for fertigation, a conventional system may either approve or reject based on fixed contaminant thresholds. IGUANA can produce a multi-layered response that: (i) contextualizes water quality relative to crop sensitivity, (ii) highlights nutrient benefits and potential risks (e.g., CEC or salinity), (iii) suggests mitigation strategies (e.g., blending and irrigation methods), and (iv) prompts coordination with water providers or regulators. Hence, IGUANA strengthens the transition from wastewater as a liability to a managed resource, supporting interdisciplinary decision-making, while preserving the integrity of stakeholder interactions and reducing the risk of biased or fragmented recommendations.

4.0.3 Financial Services and Algorithmic Risk Assessment

GenAI is increasingly utilized for investment advising, customer guidance, and automated risk disclosure. LLMs trained on historical financial corpora tend to reflect the demographic and geographic biases embedded in those datasets; for example, they may systematically deprioritize credit risk information relevant to small-business applications from underrepresented regions [2]. Current guardrails in financial recommendation systems make their bias profiles difficult to audit without generating large, structured test datasets [19]. In high-frequency financial generation environments, where advisory content is synthesized for hundreds of concurrent sessions, a blocking safety classifier of a guardrail adds non-trivial per-token latency and significant throughput degradation [20]. IGUANA’s asynchronous telemetry architecture ensures that safety evaluations run in parallel with inference, avoiding sequential bottlenecks.

4.0.4 Legal and Governmental Decision Support

The application of generative models and automated reasoning within the criminal justice system exposes an intersection of algorithmic bias and civil liberties. Predictive algorithms used in sentencing and parole decisions have demonstrated racial and socioeconomic skews, often falsely penalizing marginalized demographics due to historically biased training corpora [21]. As these systems evolve from structured tabular classifiers into autoregressive language models capable of drafting sentencing rationales or summarizing case law, the risk of bias increases. Static guardrails enforce arbitrary blocklists that fail to account for the nuanced legal context of a query, thus preventing equitable legal analysis. IGUANA mitigates this by mathematically rebalancing generation probabilities without imposing rigid blockades, thus providing accessible and fair AI assistance.

5 Discussion

This section evaluates the architectural implications of decoupling safety verification from the inference path, with a focus on runtime concurrency and statistical debiasing. We discuss the advantages of out-of-band telemetry monitoring and compare logit-level soft correction with alternative weight-mutation and pruning strategies.

5.1 The case for asynchronous and adaptive guardrails

In traditional NN architectures, the generation pipeline is halted while the guardrail processes the input prompt or the generated output sequence. These static guardrails evaluate content in isolation from generation context, user trust state, and the statistical confidence of the generating model at each step. This produces two failure modes: they over-restrict legitimate expert queries in high-trust contexts, and they under-restrict contextually dangerous outputs when the generating model is statistically uncertain but not overtly policy-violating. They apply universal safety thresholds indiscriminately, resulting in high rates of selective refusal for complex but benign topics [10]. IGUANA conceptualizes guardrails as deeply integrated asynchronous systems. By processing the Shannon entropy of token probabilities ($H(P)$) in parallel, IGUANA achieves near-zero computational latency overhead on the generation thread, incurring only a nominal 1.8 ms IPC transit delay per token. Furthermore, by replacing binary refusal blocks with dynamic statistical corrections, IGUANA mitigates the “all-or-nothing” selective refusal trap. This preserves conversational continuity for benign queries that trigger false positives under conventional static rules, while maintaining the safety envelope without interrupting the connection.

IGUANA guardrails build on fault-tolerant, high-concurrency messaging systems enabled by Erlang and its OTP [14]. The Elixir ecosystem (which runs on the Erlang VM) has introduced numerical computing libraries, such as Nx, to bridge the gap between functional concurrency and ML. However, these existing frameworks primarily focus on parallelizing the matrix multiplications in the NN’s forward pass. Rather than attempting to rewrite the core PyTorch or TensorFlow inference engines in Erlang, IGUANA isolates Erlang solely to the telemetry and safety domains. By using the Erlang supervisor swarm as an out-of-band monitoring layer via ErlPort, IGUANA achieves sub-millisecond message passing without disrupting the highly optimized GPU-bound Python ecosystems that perform LLM generation.

5.2 The case for unbiased guardrails

Prior evaluations of posterior-based guardrails (such as Trust and Safety classifiers and semantic refusal pipelines) have demonstrated that these filters inherit cognitive biases from their training data, thereby perpetuating discriminatory

decision-making in the high-stakes contexts they are designed to protect [22]. IGUANA mitigates the shortcomings of these guardrails by decoupling safety telemetry from inference, providing a real-time bias-correction mechanism derived from the generating model’s token probability distribution.

When data is limited or highly skewed, static NNs can easily overfit to biases in the training set. Current methodologies address this through highly adaptive mechanisms deployed across preprocessing, in-processing, and post-processing stages. Frameworks like A2 dynamically generate “bias-conflict samples” to counteract dataset skewness, enabling models to learn debiased representations without excessive human supervision [16]. At the structural level, solutions such as the SkewPNN introduce specialized skew-normal distribution kernels directly into the mathematical architecture to accurately model skewed distributions [15]. While runtime structural modifications (such as altering attention weights or pruning subnetworks during a forward pass) risk causing tensor shape mismatches and memory fragmentation, IGUANA avoids these complexities by confining its intervention strictly to logit-level soft correction. By performing element-wise tensor addition directly on the unnormalized log-probabilities (the `scores` tensor) pre-softmax, IGUANA dynamically steers generation without mutating active model weights or attention matrices, thereby preserving structural integrity and keeping execution overhead minimal. IGUANA utilizes these adaptive layer techniques to deploy independent adaptivity supervisors that dynamically inject bias-conflict weights based on the statistical skew of the ongoing inference context.

IGUANA treats trust initialization as a responsibility of the deploying application layer rather than the guardrail. The mechanism by which a user’s domain expertise is attested varies substantially across deployment contexts: a clinical telehealth platform may derive trust from authenticated professional credentials, a consumer wellness application may derive it from an age verification gate, and a financial advisory system may derive it from organizational role metadata. Encoding any single attestation protocol directly into the guardrail framework would impose unnecessary domain-coupling and reduce the portability of IGUANA across heterogeneous deployment environments.

6 Conclusion

This paper proposes IGUANA, an adaptive framework for integrated guardrails to enable unbiased NN architectures. IGUANA decouples safety evaluation from the sequential forward pass of generative NNs. IGUANA eliminates the sequential throughput bottleneck imposed by synchronous guardrail execution and complements binary veto thresholds with contextually adaptive responses that mitigate bias without unnecessary selective refusal. IGUANA contains a decentralized OTP supervision tree comprising a 10-actor swarm managed by the `pg` module. This swarm concurrently evaluates the Shannon entropy $H(P)$ of each autoregressive token distribution alongside a BEAM process, exerting no blocking influence on the Python inference engine. IGUANA complements the binary veto threshold with the SkewPNN adaptive correction mechanism to steer the output distribution away from unsafe content through soft-logit adjustment. IGUANA leverages Erlang’s fault-tolerant concurrency model coupled with fair and adaptive techniques. By combining decentralized orchestration, dynamic contextual thresholding, and SIMD-level compiler optimization, IGUANA provides a portable architecture for unbiased and low-latency GenAI safety.

As future work, we plan to integrate the IGUANA software acceleration kernel with high-density GPU clusters to study the scaling limits of decentralized telemetry across petascale model architectures. We aim to incorporate an explicit session-based role-attestation handshake at the application-layer boundary. Before initiating a generation session, the application would prompt the user to declare their professional relationship to the subject matter through a short, structured onboarding exchange. The declared role would be mapped to a normalized trust score on the continuous interval $[0.0, 1.0]$ and passed as a discrete step during session initialization. This layered separation maintains IGUANA as a domain-agnostic guardrail while providing a concrete deployment pathway for context-sensitive adaptiveness.

Acknowledgments: This work is funded by the EU NextGeneration under the Juraj Dobrila University of Pula institutional research project number IIP_UNIPU_010162.

References

- [1] S. R. Thorat and D. G. Jha, “Revolutionizing healthcare: The transformative power of generative ai—risks and rewards,” in *Technologies for Energy, Agriculture, and Healthcare*. CRC Press, 2025, pp. 57–67.

- [2] S. Gehrmann, C. Huang, X. Teng, S. Yurovski, A. Bhorkar, N. Thomas, J. Doucette, D. Rosenberg, M. Dredze, and D. Rabinowitz, “Understanding and mitigating risks of generative ai in financial services,” in *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, 2025, pp. 2570–2586.
- [3] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020.
- [4] T. B. Brown *et al.*, “Language models are few-shot learners,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
- [5] J. De Freitas and I. G. Cohen, “The health risks of generative ai-based wellness apps,” *Nature medicine*, vol. 30, no. 5, pp. 1269–1275, 2024.
- [6] T. Rebedea, R. Dinu, M. N. Sreedhar, C. Parisien, and J. Cohen, “Nemo guardrails: A toolkit for controllable and safe llm applications with programmable rails,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2023, pp. 431–445.
- [7] M. Weber, M. Huber, M. Auch, A. Döschl, M.-E. Keller, and P. Mandl, “Digital guardians: Can gpt-4, perspective api, and moderation api reliably detect hate speech in reader comments of german online newspapers?” in *2025 38th Conference of Open Innovations Association (FRUCT)*. IEEE, 2025, pp. 318–326.
- [8] Q. Zhang, Z. Xiong, and M. Mao, “Llm safeguard is a double-edged sword: Exploiting false positives for denial-of-service attacks,” in *Proceedings of the 2025 Workshop on Large AI Systems and Models with Privacy and Security Analysis*, 2025, pp. 1–10.
- [9] L. Goukassian, “Ternary moral logic (tml) and the future of ai governance: A technical analysis for nvidia,” *Available at SSRN 5856362*, 2025.
- [10] B. Wang, W. Chen, H. Pei, C. Xie, M. Kang, C. Zhang, C. Che, U. Gao, B. Hou *et al.*, “Decodingtrust: A comprehensive assessment of trustworthiness in gpt models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [11] A. Khorramrouz and S. Levy, “Characterizing selective refusal bias in large language models,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 11 305–11 326.
- [12] F. Jia, Z. Ye, S. Lai, K. Shu, J. Gu, A. Bibi, Z. Hu, D. Jurgens, J. Evans, P. H. Torr *et al.*, “Can large language model agents simulate human trust behavior?” *Advances in neural information processing systems*, vol. 37, pp. 15 674–15 729, 2024.
- [13] O. Shaikh, H. Zhang, W. Held, M. Bernstein, and D. Yang, “On second thought, let’s not think step by step! bias and toxicity in zero-shot reasoning,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- [14] J. Armstrong, “Making reliable distributed systems in the presence of software errors,” Ph.D. dissertation, Royal Institute of Technology, Stockholm, 2003.
- [15] S. M. Naik, T. Chakraborty, M. Panja, A. Hadid, and B. Chakraborty, “Skew-probabilistic neural networks for learning from imbalanced data,” *Pattern Recognition*, vol. 165, p. 111677, 2025.
- [16] J. An, T. Kim, D. Ko, S. Lee, and S. S. Woo, “A[^] 2: Adaptive augmentation for effectively mitigating dataset bias,” in *Proceedings of the Asian Conference on Computer Vision*, 2022, pp. 4077–4092.
- [17] Y. Zhao, H. Wang, Y. Liu, W. Suhuang, X. Wu, and Y. Zheng, “Can LLMs replace clinical doctors? exploring bias in disease diagnosis by large language models,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.
- [18] K. T. Cheng, A. L. Gentile, P. Li, C. DeLuca, and G.-J. Ren, “Don’t be my doctor! recognizing healthcare advice in large language models,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2024.

- [19] M. A. Beltran, M. I. Ruiz Mondragon, and S. H. Han, “Comparative analysis of generative ai risks in the public sector,” in *Proceedings of the 25th Annual International Conference on Digital Government Research*, 2024, pp. 610–617.
- [20] Y. Leviathan, M. Kalman, and Y. Matias, “Fast inference from transformers via speculative decoding,” in *International Conference on Machine Learning (ICML)*, 2023.
- [21] S. A. Zottola, S. L. Desmarais, E. M. Lowder, and S. E. Duhart Clarke, “Evaluating fairness of algorithmic risk assessment instruments: The problem with forcing dichotomies,” *Criminal Justice and Behavior*, 2021.
- [22] J. M. Echterhoff, Y. Liu, A. Alessa, J. McAuley, and Z. He, “Cognitive bias in decision-making with LLMs,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.